

Redmi 6 Pro Level 3 Maintenance Guide V01

: Redmi 6 Pro Level 3 Maintenance Guide V01

Scope of application:

Change history of analysis center,
motherboard and whole machine
maintenance factory:

First edition 2018/7/18

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1. Introduction of basic information

1.1 Product Overview

Product Overview:



红米6 Pro

AI双摄 高颜值大电量

5.84" 异形全面屏
19:9 FHD+, 高屏占比

后置1200万+500万
AI 双摄, 随手拍大片

骁龙625 八核处理器
14nm 制程, 强劲性能

4000mAh大电量
畅快使用, 精彩不断

AI人脸解锁
单反般背景虚化

2+1 卡槽
最高可扩展256GB



红米6 Pro

3GB + 32GB / 4GB + 32GB / 4GB + 64GB
骁龙 625 八核处理器, 最高主频 2.0GHz
AI双摄相机1200万+500万, 500万前置美颜自拍、人脸解锁
高度: 149.33mm 宽度: 71.68mm 厚度: 8.75mm 重量: 178 g
5.84 英寸 (对角线) 19:9 全面屏 2280x1080分辨率, 432 PPI, 对比度1500:1, 500 (typ) nit 亮度
支持双Nano-Sim卡+micro-SD扩展, 全网通3.0, 任意卡槽均可设置为主卡支持双卡不限运营商, 不能同时支持双卡4G驻网
4000mAh (typ) / 3900mAh (min) 5V/2A
支持802.11a/b/g/n WiFi协议, 支持 2.4G WiFi / 5G WiFi / WiFi Direct / WiFi Display, 蓝牙4.2 无线技术
GSM (频段 B2/3/5/8), CDMA1X / EVDO (频段 BC0), WCDMA (频段B1/2/5/8), TD-SCDMA (频段 B34/39), TDD-LTE (频段 B34/38/39/40/41 (100MHz), FDD-LTE (频段 B1/3/5/7/8)

1.2 Red 6 Pro special welding fixture

Material code: SCNC020057200

1.3 Red 6 Pro power supply adapter

Material code: SCNC030025900 (common to Xiaomi 5X power supply adapter)

1.4 Pasting position and specification of maintenance label

The pasting position of the maintenance label is as shown in the figure:



1.5 Matters needing attention in motherboard maintenance

1. Before soldering the motherboard, the heat dissipation silica gel needs to be removed to prevent the chip squeezed under the heat dissipation silica gel pad from causing motherboard failure during heating.
2. Interface and sim card socket are plastic components, and anti-welding measures should be taken when welding around these components.
3. U1500 has a binding relationship with U100. When U1500 is damaged, new materials can be replaced separately. If U100 is replaced, a new U1500 needs to be replaced.

1.6 Brush machine type

Brush platform: MiFlash

Whole machine special factory package:

Fastboot factory special brush machine

mode: brush machine in Fastboot mode

Special factory package for changing

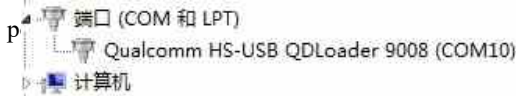
number: (keep NV package) Special

factory package for maintenance:

(clear NV package) Brushing mode:

deep brushing (EDL)

1. Short-circuit TP2410 and TP2707 when the mobile phone is turned off, so that the mobile phone enters deep brush mode (identify



- 2 Select the corresponding software package, check "Delete All" at the bottom of MiFlash, click "Load Device", and click "Brush Machine".



After the factory package is used to enter the factory version in Fastboot mode, BL lock needs to be unlocked before brushing, and operations such as writing numbers can be carried out normally after brushing. EDL package (EDL refers to emergency brushing machine, that is, deep brushing machine) special package for deep brushing machine, which can only be used if it is shorted into deep brushing mode, and does not need to unlock BL lock before brushing machine. When starting up for the first time after brushing EDL package, it is advanced FFBM mode (fast factory boot mode) and selects "Item test" for relevant tests.

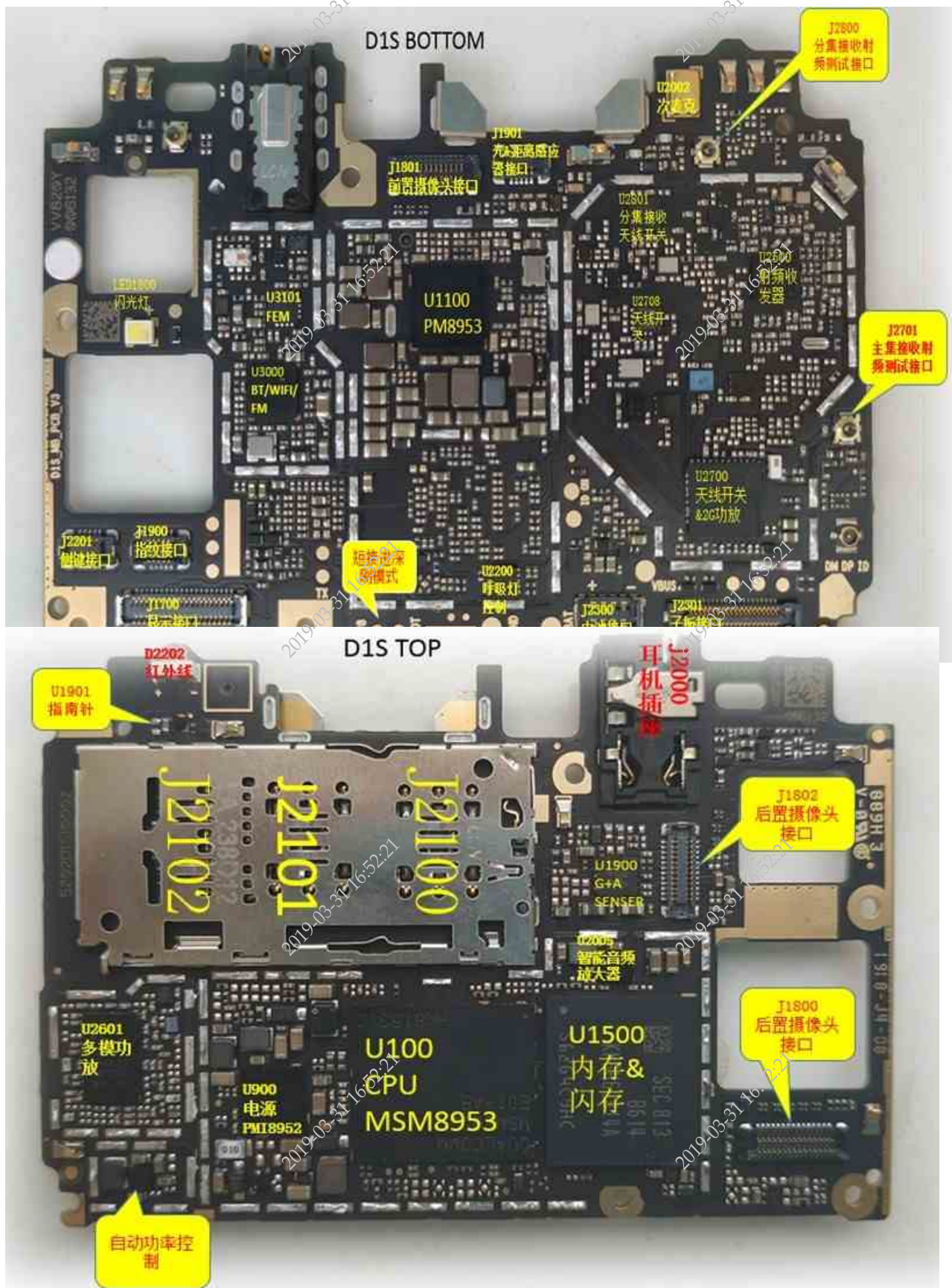


1.7 RF calibration test correlation

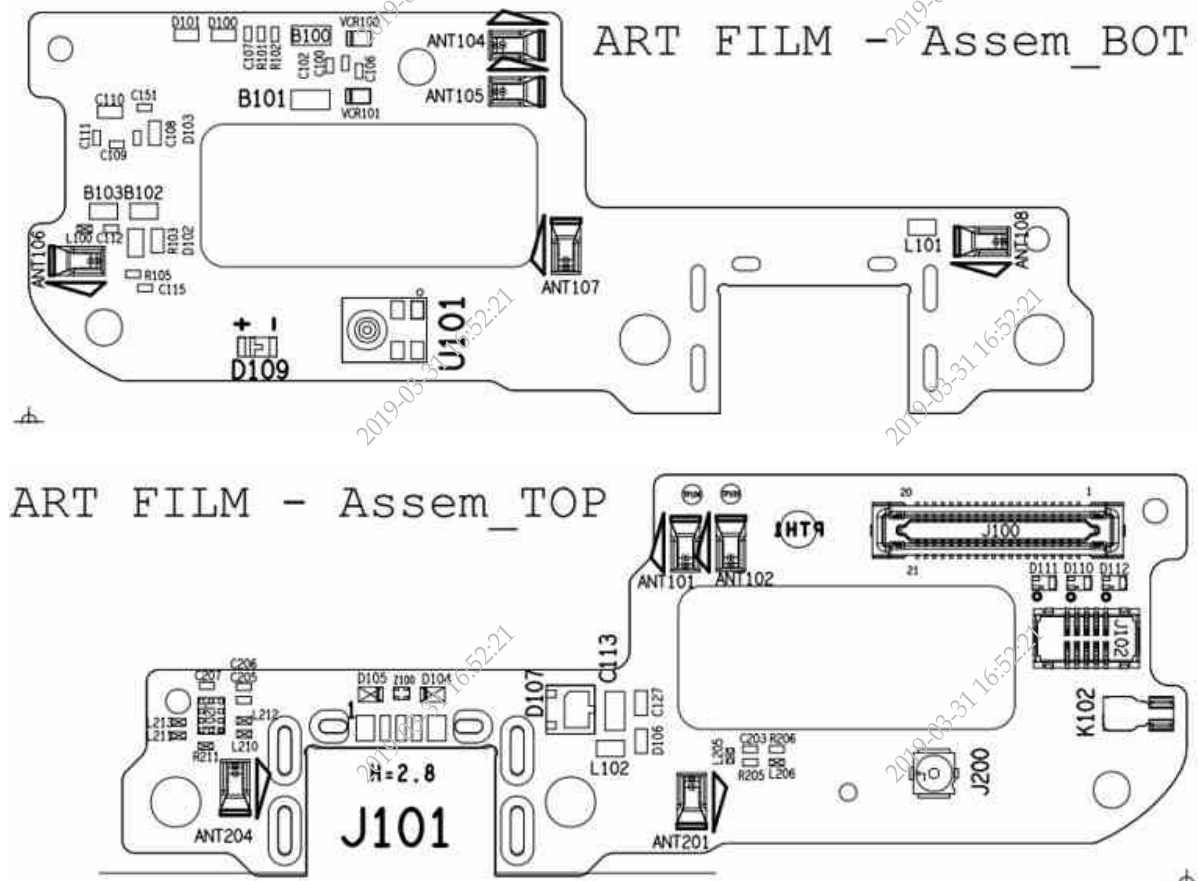
1. Deep brush factory software (Redmi 6 Pro-easeNV-EDL), calibrated after writing FSN with DT tool. Note that the DT tool needs to be updated to DT 2018.7.12.0_factory or later, please follow the instructions.
2. Open the calibration software, and connect the main receiving antenna and the auxiliary receiving two RF lines when the mobile phone is turned on. After connecting USB cable, the mobile phone starts RF calibration.
3. After successful calibration, write the flag bits with Xiaomi 5X's flag bit writing tool: "NV Read & Write V2.2". After writing the flag bit successfully, "successes" will be displayed on the tool. After the calibration is completed, you can see that BT and FT display "FAIL" in the version information in factory mode. Only when the flag bit is written successfully will BT and FT display "PASS" in the version information in factory mode.
4. If you want to test the signal function after successful calibration and writing the flag bit, you need to use DT tool to restart the phone.

2 Introduction to Motherboard Modules

2.1 Distribution map of red 6 Pro motherboard components



2.2 Distribution map of red 6 Pro panel components



2.3 Brief introduction of startup timing and key signal measurement table of Red 6 Pro

Boot timing diagram:

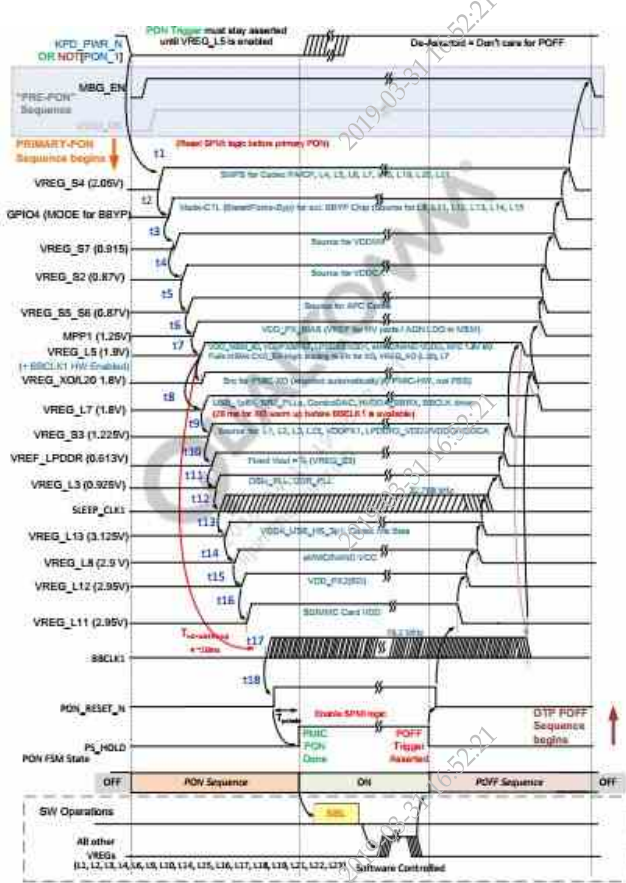


Figure 3-5 PM8953 power-on sequence

Boot Timing:

开机时序测量表		
Symbol	测量值	测量点
KYPD_PWR_N1	1.8V	R2204
VREG_S4_2P05	2.05V	R1106
VREG_S7_0P915	0.915V	C601
VREG_S2_0P8625	0.8625V	C607
REG_S5_S6_0P8625	0.8625V	C655
VREG_L5_1P8	1.8V	C652
VREG_XO	1.8V	C1105
VREG_L7_1P8	1.8V	C1305
VREG_S3_1P225	1.225V	R1300
VREF_LPDDR3	0.6V	R1513
VREG_L3_0P925	0.925V	C700
SLEEP_CLK1	32K	R1105
VREG_L13_3P075	3.075V	C1311
VREG_L8_2P9	2.9V	C1306
VREG_L12_VDDPX2_SDC	2.95V	C638
REG_L11_2P95	2.95V	C1309
BBCLK1	19.2MHZ	R1108
PON_RESET_N	1.8V	C100
PS_HOLD	1.8V	TP2413

low signals are also signals in startup timing):

CPU供电测量表		
Symbol	测量值	测量点
VREG_S2_OP8625	0.8625V	C608
VREG_S7_OP915	0.915V	C601
VREG_S5_S6_OP8625	0.8625V	C655
VREG_S3_1P225	1.225V	C621
VREG_L13_3P075	3.75V	C704
VREG_L3_OP925	0.925V	C700
VREG_L5_1P8	1.8V	C1009
VREG_L7_1P8	1.8V	C1305
VREG_L6_1P8	1.8V	C3005
VREG_S1_OP8625	0.8625V	C721
VREG_L12_VDDPX2_SDC	2.9V	C638
VREG_L14_UIM1	1.8V (插卡测)	C650
VREG_L15_UIM2	2.9V (插卡测)	C651
VREG_L19_1P3	1.3V	C1315
VDDA_GPS_BBRX	1.2V	C713
VDD_PX_BIAS_MPP_1	1.2V	C606

U1100 chip bitmap:

1 GND_XD_IS O	2 XTAL 18M_IN	3 XTAL 18M_OUT	4 VREG_XD	5 VREG_RFC LK	6 VREG_L9	7 VREG_L7	8 VREF_LPD OR	9 VREG_L1	10 GND_S4	11 VSW_S4	12 VDD_S4	13 VDD_S4	
14 VCOIN	15 GND_XO_IS O	16 GND_XO	17 GND_RFCL K	18 VREG_L18	19 BB_CLK2	20 VREG_L27	21 VDD_L4_5 _6_7_18_19	22 VREG_L8	23 VREG_L19	24 GND_S4	25 VSW_S4	26 VDD_S4	27 PON_RES T_N
28 VREG_L12	29 VDD_L8_11 _12_13_14_15	30 VREG_L8	31 VDD_XO_R FCLK	32 BB_CLK1	33 VDD_L9_10 _17_18_22	34 VREG_L17	35 VDD_L4_5 _6_7_18_19	36 VDD_L1	37 SPM_DATA	38 VREG_S4	39 GND_S1	40 GND_S1	
41 RF_CLK1	42 VREG_L11	43 RF_CLK2	44 VDD_L8_11 _12_13_14_15	45 VREG_L14	46 VX_CLK3	47 VREG_L10	48 VREG_L16	49 VREG_L5	50 VREG_L4	51 SPM_CLK	52 PS_HOLD	53 VSW_S1	54 VSW_S1
	55 BB_CLK1_E N		56 VREG_L13	57 REF_BYP	58 VREG_L15	59 VPH_PWR		60 AVDD_BYP	61 DVDD_BYP	62 VREG_S1	63 KPD_PWR_N	64 VDD_S1	65 VDD_S1
66 LINEOUT_M	67 LINEOUT_P	68 HPH_L	69 MIC_BIAS1	70 GND_REF	71 GND	72 GND_LOAD C	73 XO_THERM	74 PA_THERM	75 VREG_S2	76 RESIN_N	77 PON_1	78 GND_S2	79 GND_S2
80 VNEG_HPH	81 VDD_HPH	82 HPH_REF	83 MIC_BIAS2	84 GND_CFLT	85 GND	86 GND	87 GND	88 GND	89 SLEEP_CLK 1	90 GPIO_6	91 VSW_S2	92 VSW_S2	
93 EARD_F	94 EARD_M	95 HPH_R	96 MIC1_IN_P	97 MIC1_IN_M	98 GND	99 GND	100 GND	101 OC1	102 GND	103 VREG_S7	104 GPIO_5	105 VDD_S2	106 VDD_S2
107 VDD_SPKR _PA		108 MIC2_IN	109 HS_DET	110 MPP_3	111 VBT_1	112 MPP_1	113 VCOIN	114 OC1	115 VREG_S5	116 VREF_NEG _S5	117 GPIO_4	118 GND_S7	119 GND_S7
120 SPKR_DRV _F	121 SPKR_DRV _M		122 MIC3_IN	123 OPT_2	124 CBL_PWR _N	125 MPP_2		126 VDD_L23	127 VREG_S6	128 VREF_NEG _S6	129 GPIO_8	130 VSW_S7	131 VSW_S7
132 GND_SPKR _PA	133 CP_VNEG	134 PDM_RX0 _DRE	135 PDM_RX0	136 PDM_SYNC	137 VREG_S3	138 VREG_L23	139 VDD_L2_3	140 VREG_L2	141 GPIO_2	142 GPIO_3	143 GPIO_7	144 VDD_S7	145 VDD_S7
146 CP_C1_M	147 GND_CP	148 PDM_RX1 _DRE	149 PDM_TX	150 PDM_CLK	151 MPP_4	152 GPIO_1	153 VREG_L3	154 VDD_S6	155 VSW_S6	156 GND_S6	157 VDD_S5	158 VSW_S5	159 GND_S5
160 CP_C1_P	161 VDD_CP	162 BOOST_SN S	163 VDD_AUDIO _IO	164 PDM_RX2	165 VDD_S3	166 VSW_S3	167 GND_S3	168 VDD_S6	169 VSW_S6	170 GND_S6	171 VDD_S5	172 VSW_S5	173 GND_S5
174 GND_BOOS T	175 GND_BOOS T	176 VSW_BOOS T	177 VREG_BOO ST	178 PDM_RX1	179 VDD_S3	180 VSW_S3	181 GND_S3	182 VDD_S6	183 VSW_S6	184 GND_S6	185 VDD_S5	186 VSW_S5	187 GND_S5

Audio

Configurable I/Os

Do not connect

General housekeeping

Ground

IC-level interfaces

Input power management

Output power management

Power

1 VDS_N_OUT	2 VSW_DIS_N	3 VDD_DIS_N	4 GND_DIS_P	5 VDS_P_OUT	6 VREG_WLED	7 VDD_WLED	8 GND_WLED	9 VSW_WLED	10 MPP_3	11 HAP_PWM_IN	12 HAP_PWM_IN
13 VDS_N_OUT	14 VSW_DIS_N	15 DIS_N_CAP_REF	16 VSW_DIS_P	17 VSW_DIS_P	18 NC	19 NC	20 WLED_SINK2	21 WLED_SINK1	22 MPP_1	23 GNDC	24 HAP_OUT_N
25 VDD_1PB_DIS_N	26 GND_DIS_N_REF	27 VDS_N_FB	28 VDD_DIS_P	29 VDS_P_FB	30 WLED_CABC	31 GND_WLED_1	32 WLED_SINK3	33 WLED_SINK4	34 MPP_2	35 HAP_OUT_P	36 GND_HAP
37 GNDC	38 SPMI_CLK	39 SPMI_DATA	40 NC	41 NC	42 VDD_MSAT_IO	43 BUA	44 NC	45 GNDC	46 VREG_ADC_LDO	47 MPP_4	48 VDD_HAP
49 GNDC	50 CLK_IN	51 DIS_SCTRL	52 GNDC	53 AVDD_BYP	54 DVDD_BYP	55 SHDN_N	56 GNDC	57 VDD_ADC_LDO	58 REF_BYP	59 GND_REF	60 VDD_TORCH
61 CS_PLUS	62 BATT_PLUS	63 GPIO_2	64 GNDC	65 GNDC	66 GNDC	67 GNDC	68 RESIN_N	69 PS_HOLD	70 GNDC	71 FLASH_LED1	72 VDD_FLASH
73 CS_MINUS	74 BATT_MINUS	75 GPIO_1	76 GNDC	77 GNDC	78 GNDC	79 GNDC	80 USB_ID	81 WIPWR_DIV2_EN	82 GNDC	83 FLASH_LED2	84 VDD_FLASH
85 R_BIAS	86 BATT_ID	87 GNDC	88 DC_EN	89 WIPWR_RST_N	90 USB_ID_RVAL1	91 CHG_LED	92 USB_SNS	93 SYSON	94 FLASH_OUT	95 FLASH_OUT	96 FLASH_OUT
97 BATT_THERM	98 VAA_CAP	99 GNDC	100 USB_EN	101 USB_CS	102 CHG_EN	103 PGOOD_SYSON	104 DC_SNS	105 BOOT_CAP	106 VSW_CHG	107 VSW_CHG	108 VSW_CHG
109 GNDC	110 GND_FG	111 USB_DP	112 USB_ID_RVAL2	113 CHG_OUT	114 VPH_PWR	115 GNDC	116 VSW_CHG	117 VSW_CHG	118 VSW_CHG	119 VSW_CHG	120 VSW_CHG
121 GNDC	122 GND_REF_CHG	123 USB_DM	124 VBAT_SNS	125 CHG_OUT	126 CHG_OUT	127 VPH_PWR	128 GND_CHG	129 GND_CHG	130 USB_MID	131 USB_MID	132 USB_MID
133 GNDC	134 KYPD_PWR_N	135 V_ARB	136 GNDC	137 CHG_OUT	138 VPH_PWR	139 VPH_PWR	140 WIPWR_CHG_OK	141 GND_CHG	142 USB_IN	143 USB_IN	144 USB_IN

Configurable IOs

General Housekeeping

Ground

IC-level Interfaces

Input Power Management

No Connect

Power

User Interfaces

3 Troubleshooting

3.1 1 on/off failure

In the process of maintenance without starting up, pay attention to observe whether the motherboard components are damaged, breakdown, liquid inflow, etc. Then follow the principle of software before hardware, and measure according to the starting time sequence when measuring concretely.

3.1.1 Constant current without startup

Analysis ideas:

1. Software upgrade, troubleshooting software.
2. If the current is still constant after brushing, measure whether the startup timing and CPU working conditions are normal.
3. If the brush machine reports an error, measure whether the working conditions of U100 and U1500 are normal.

Maintenance Case 1

Fault phenomenon:

60mA constant current

fault cause: software failure

Maintenance analysis: Repair the fault after cleaning the NV factory package deeply.

Maintenance Case 2

Fault phenomenon: 60mA

constant current fault

cause: U100

Maintenance analysis: The deep brush factory package can be brushed, but the fault phenomenon remains unchanged, the measured voltage and working conditions are not abnormal, and the fault is repaired after replacing U100.

3.1.2 No startup current is not maintained

Analysis ideas:

1. Software upgrade and troubleshooting.
2. Measure whether the USB signal line is normal (USB_ID; USB_DM; USB_DP).
3. Measure whether the startup timing signal is normal.

4. Measure whether the U100 power supply is normal.

Maintenance Case 1

Failure phenomenon: 40mA
does not maintain failure
cause: software failure

Maintenance analysis: There is no abnormality in appearance, and the fault is repaired after deep brushing of NV factory package.

Maintenance Case 2

Fault phenomenon: 60mA
does not maintain faulty
component: U100

Maintenance analysis: Try to brush the NV factory package and report the error, and repair the fault after replacing U100.

3.1.3 No startup current

Analysis ideas:

1. Check whether the appearance of J2300 and J2201 is damaged, and if the appearance is normal, measure whether the ground value of J2300 is normal (BAT_ID, BAT_THERM, VBATT).
2. Power up to measure whether the VPH_PWR output is normal and whether the KYPD_PWR_N1 (R2204) 1.8 V voltage is normal. If there is no output, replace U900.
3. Measure whether the startup timing signal is normal.

Maintenance Case 1

Fault phenomenon: startup
current 0mA Fault
component: U1100

Maintenance analysis: Press the startup key, the power meter has no response, KYPD_PWR_N1 has 1.8 V voltage, VREG_S4_2P05 has no voltage output according to the startup time sequence, and the fault is repaired after replacing U1100.

3.1.4 Leakage without starting up

Maintenance analysis ideas:

1. First of all, visually check whether there are components rupture, breakdown, liquid corrosion and discoloration in the appearance of the motherboard.
2. Measure whether VBATT and VPH_PWR are short-circuited. If both VBATT and VPH_PWR are short-circuited, find VBATT short-circuited elements first and then VPH_PWR short-circuited elements.
3. Measure whether there is short circuit in other power supply lines, and if there is short circuit in signal, find out the faulty components in its path.
4. Power on to find heating elements or quickly find them with the help of thermal imaging instruments.

Maintenance Case 1

Fault Phenomenon:
Leakage 3A Fault
Component: C2301

Analysis idea: Measure VBAT short circuit, no special heating elements, exclude the elements on its path in turn, and its path returns to normal after taking off C2301. Replace new components.

Maintenance Case 2

Fault phenomenon:
leakage 40mA Fault
component: C1010

Maintenance analysis idea: Measure VPH_PWR short circuit, power-on touch C1010 heat is obvious, and its path returns to normal after taking it off. Replace new components.

Maintenance Case 3

Fault Phenomenon:
Leakage 3A Fault
Component: C1013

Maintenance analysis idea: Measure VPH_PWR short circuit, power-on touch C1013 heat is obvious, and its path returns to normal after taking it off. Replace new components.

Maintenance Case 4

Fault Phenomenon:
Leakage 23mA Fault
Component: U900

Maintenance analysis ideas: Measure VPH_PWR short, increase the power meter voltage to the main board after the touch U900 micro-heat, take it off its path to return to normal, replace new components to repair the fault.

3.1.5 Dynamic shutdown

Maintenance analysis ideas:

1. Check whether the appearance around U1100 and U900 is normal, and give priority to measurement if there is any abnormality.
2. Detect whether the clock signal is normal.
3. Measure whether the startup timing signal is normal.
4. Check whether the battery charge detection and board-level temperature detection circuits are normal.

3.2 2 Restart failure

Analysis ideas:

1. Check whether the KYPD_PWR_N1 1.8 V voltage is pulled low.
2. Software upgrade, troubleshooting software.
3. Measure whether I2C to ground and voltage are normal.
4. Measure whether the U100 is powered and the clock is normal.
5. Measure whether there is short circuit in the power supply line of the motherboard.
6. Replace U100.

Maintenance Case 1

Failure Phenomenon:

Standby Restart

Failure Cause: U100

Maintenance analysis ideas: try to clean the NV factory package without effect, replace U100 & U1500 after the fault repair.

3.3 3 Crash failure

Analysis ideas:

1. Software upgrade, troubleshooting software.
2. Measure whether BAT_ID and BAT_THERM are normal.
3. Measure whether the startup timing is normal.
4. Measure whether the U100 power supply is normal.
5. Replace U100.

Maintenance Case 1

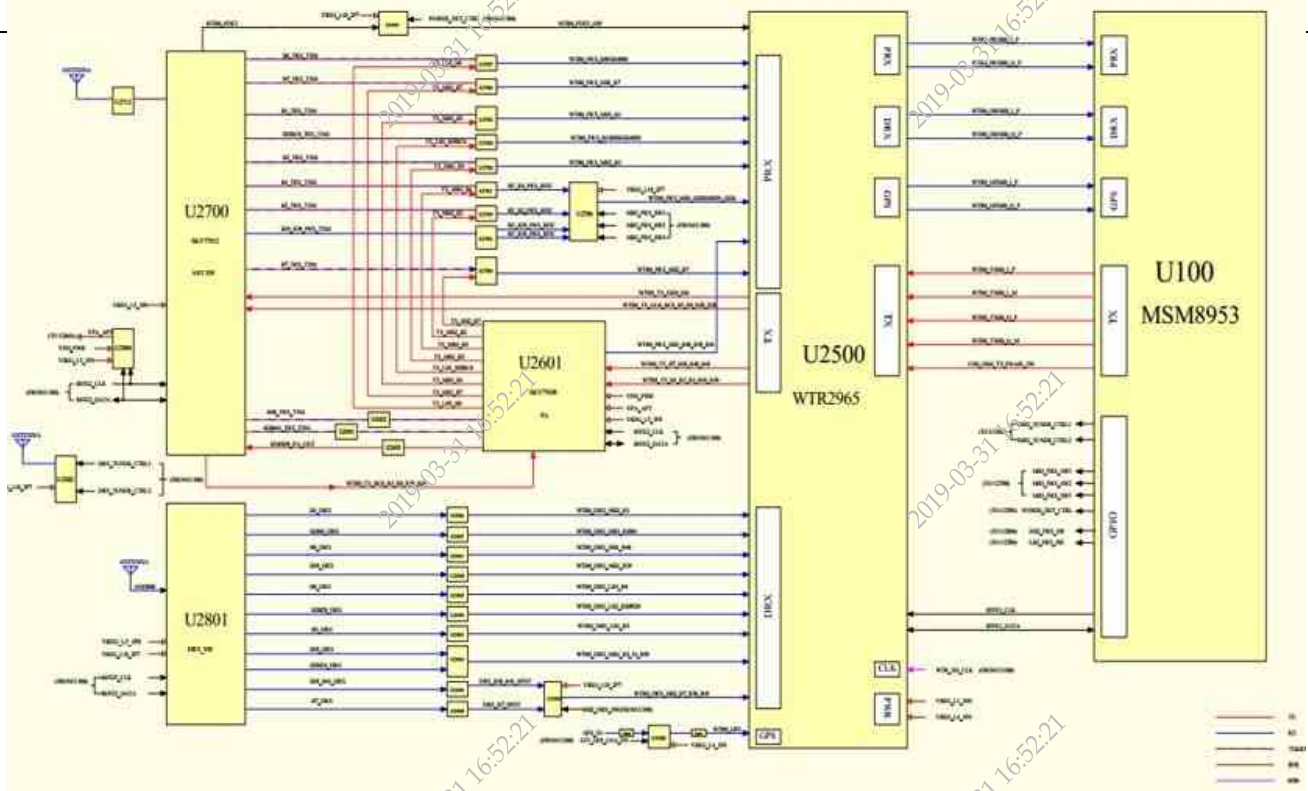
Failure phenomenon: Slow startup, crash cause: U100

Maintenance analysis idea: Brush machine is invalid, and the fault is repaired after replacing U100-U1500.

3.4 4 signal failure

The RF transceiver (U2500) of Redmi 6 Pro is a WTR2965 chip, which is highly integrated with multi-mode and multi-band RF CMOS radio receiving integrated circuit. GSM (band B2/3/5/8), CDMA1X/EVDO (band BC0), WCDMA (band B1/2/5/8), TD-SCDMA (band B34/39), TDD-LTE (band B34/38/39/40/41 (100MHz), FDD-LTE (band B1/3/5/7/8)

RF Principle Block Diagram:



RF main IC meter:

信号测量表			
元件名称	Symbo	测量值	测量点
U2500	VREG_L1_1P0	1V	R2503
	VREG_L4_1P8	1.8V	R2508
	WTR_XO_CLK	19.2MHZ	R2506
U2900	VPH_PWR	4V	C2901
	VPA_APT	4V	C2903
	VREG_L5_1P8	1.8V	C2902
	RFFE2_CLK	450	C2905
	RFFE2_DATA	450	C2906
U2601	VPH_PW	4V	C2617
	VREG_L5_1P8	1.8V	C2624
	RFFE2_CLK	450	C2626
	RFFE2_DATA	450	C2627
U2801	VREG_L18_2P7	2.7V	C2801
	VREG_L5_1P8	1.8V	C2803
	RFFE5_CLK	450	C2842
	RFFE5_DATA	450	C2802

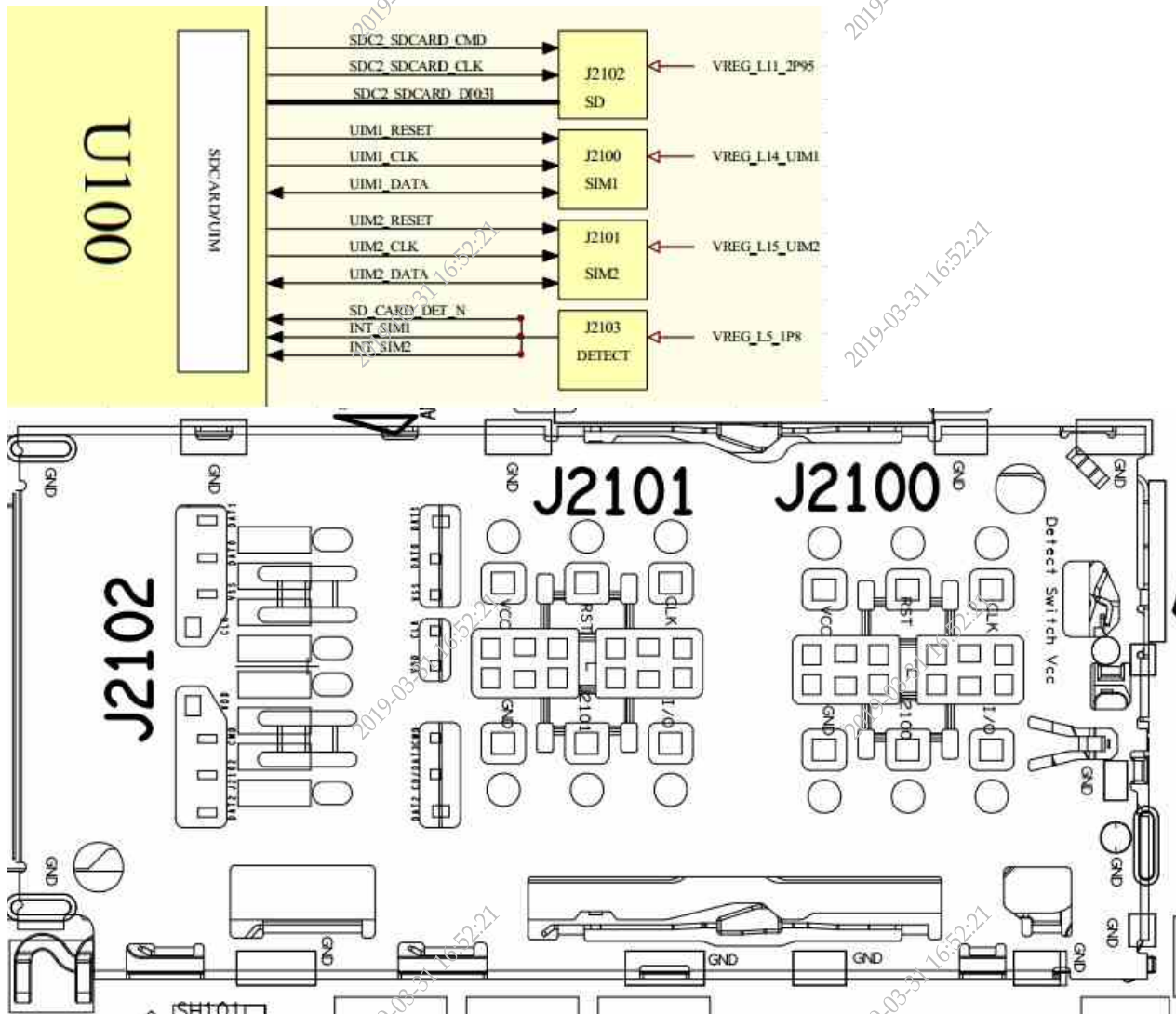
1. Insert SIM card to ensure normal recognition, and eliminate faults that do not recognize SIM card.
 2. Software upgrade, troubleshooting software.
 3. RF calibration, check the specific test items through the test report. However, measure the RF path according to the corresponding system and principle block diagram to find the fault point.
 4. Measure whether the power supply of RF circuit is normal, and whether the RFFE (1/2/5) CLK/DAT signal is normal.
 5. If the RF calibration is normal and there is still no signal, check whether the line from J3300 to J2701 is normal. Maintenance Case 1
- Fault phenomenon: No signal for card recognition
Fault cause: U2700

Maintenance analysis: Calibrate after brushing the machine, report error "Internal Device Cal" and repair the fault after replacing U2700.

3.5 5 SIM card failure

Redmi 6 Pro supports dual Nano-Sim card + micro-SD expansion, and full Netcom 3.0 supports dual cards without limited operators, and cannot support dual cards 4G resident network at the same time.

Principle block diagram of SIM card:



1. Check whether the SIM card pin is deformed, oxidized or broken, and carefully observe whether there is virtual welding phenomenon in the solder joint of SIM card holder.
2. Check whether the baseband version of the mobile phone is normal, and then check the related functions of SIM card if the baseband information is normal.
3. Measure whether the SIM card aims at the ground value normally.
4. Start up and test whether the power supply, clock, reset and data signal of SIM card are normal.
5. If the above signal is normal, replace U100.

Maintenance Case 1

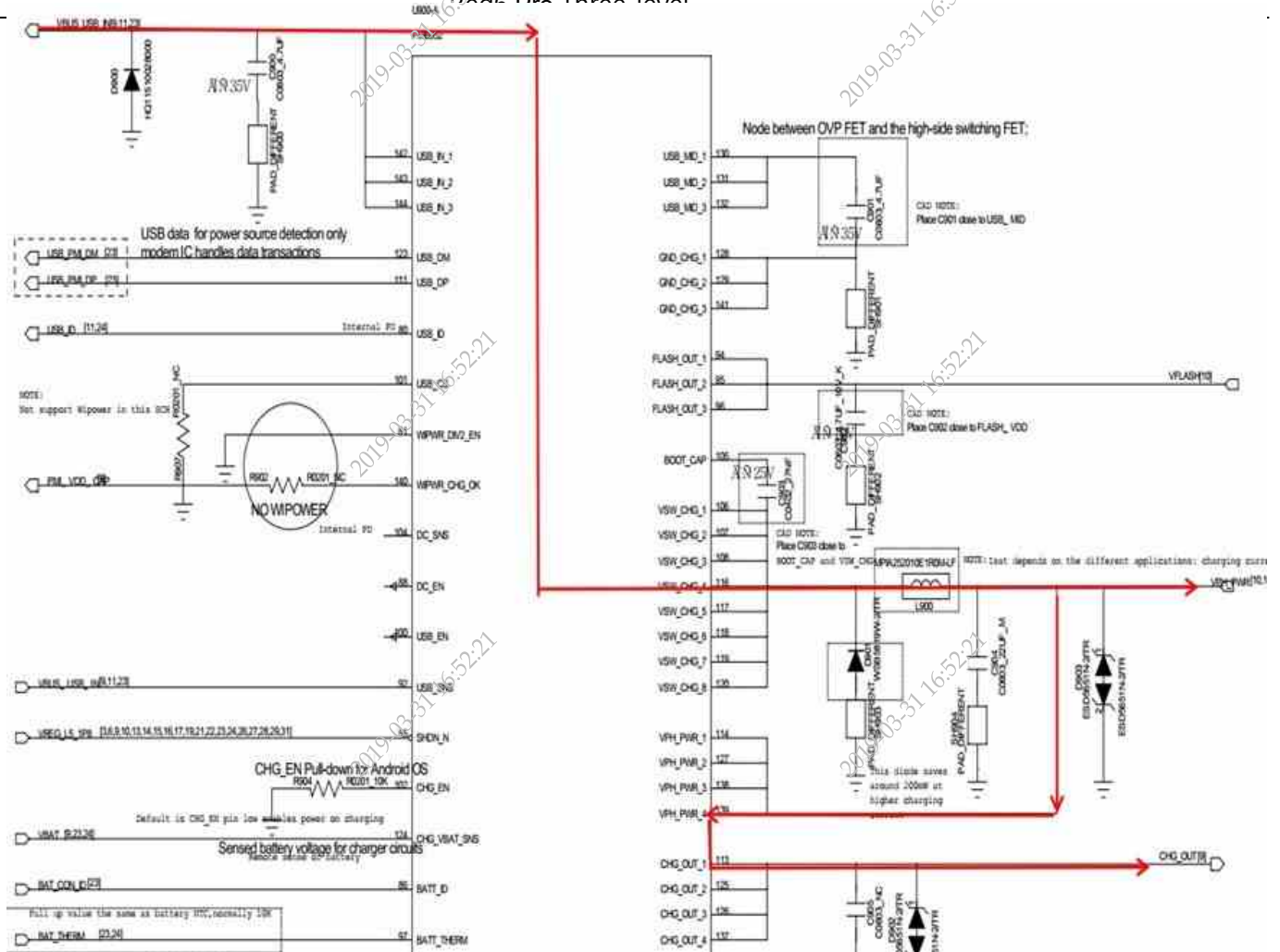
Failure phenomenon:

Card 2 does not
recognize card Failure
cause: J2101 is damaged

Maintenance analysis: Visual inspection found that J2101 card pin is deformed, and the fault is repaired after replacing new components.

3.6 6 Charging failure

Charging schematic diagram:



Analysis ideas:

1. Check whether the appearance of J2300 and J2301 is normal.
2. Measure whether VBUS_USB_IN, USB_PMI_DP and USB_PMI_DM signals are normal to ground with multimeter diode file, and measure whether VBATT, BAT_CON_ID and BAT_THERM of J2300 are normal to ground.
3. Ensure that the above signals are normal and consider replacing U900.

Maintenance Case 1

Failure

Phenomenon: No

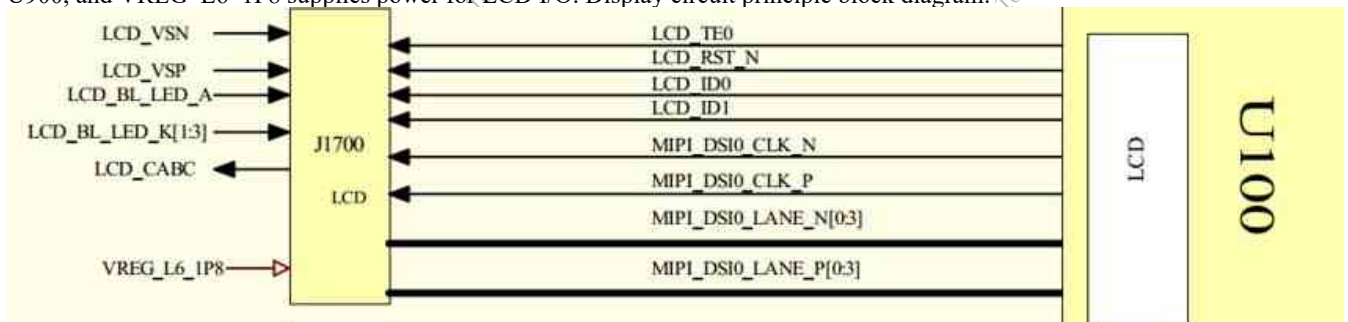
Charging Failure

Cause: D2307

Maintenance analysis: Measure the short circuit of VBUS to ground, check and find that D2307 has burnt marks, and repair the fault after replacing new components.

3.7 7 Show fault

J1700 is the connection interface of display screen, backlight driver and display driver are generated by U900, and VREG_L6_1P8 supplies power for LCD I/O. Display circuit principle block diagram:



Display some measurement tables:

显示测量表		
Symbol	测量值	测量点
LCD_VSN	-5.5V	C1007
LCD_VSP	5.5V	C1011
LCD_BL_LED_K1	0.4V	R1707
LCD_BL_LED_K2	0.4V	R1718
LCD_BL_LED_K3	0.4V	R1708
LCD_BL_LED_A	4V(息屏), 17V(亮屏)	C1703
LCD_RST_N	1.8V	R1705
LCD_CABC	1.8V	R1703

Maintenance analysis ideas:

1. Visually check whether J1700 and its peripheral components are damaged or falsely welded.
2. Brush the machine to eliminate software faults.
3. Measure whether the ground value of each pin of J1700 is normal with multimeter diode gear.

LCM CONN J1700																			
1	3	5	7	9	11	13	15	17	19	21	23	25	27	29	31	33	35	37	39
GND	455	∞	1245	1260	GND	507	GND	516	GND	1260	GND	∞	390	∞	595	595	595	∞	GND
GND	355	355	GND	355	355	GND	355	355	GND	355	355	GND	355	355	GND	∞	660	GND	∞
2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40

4. If the ground value is normal, measure whether the power supply and control signals in the "display meter" are normal.
5. Replace U100.

Maintenance Case 1

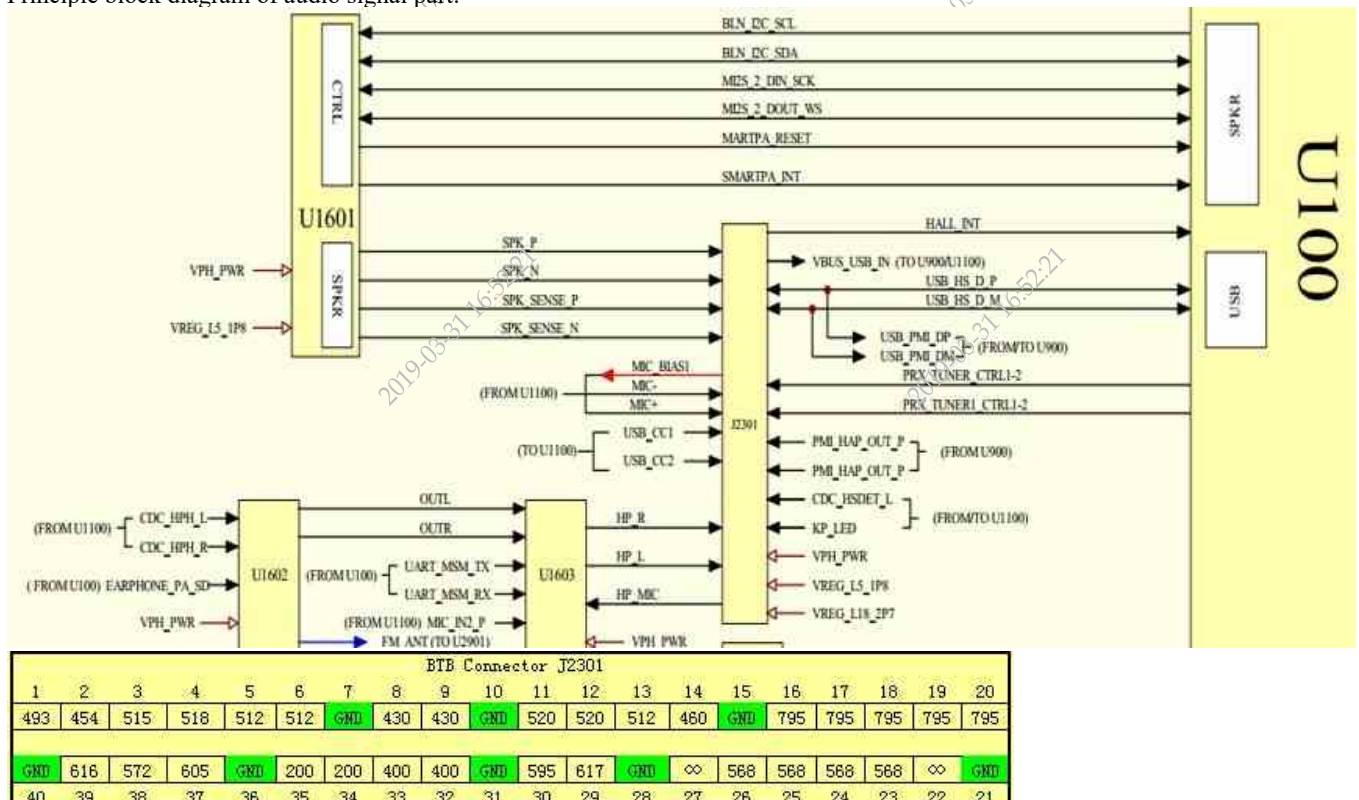
Failure phenomenon: No display after starting white rice. Failure cause: U900

Maintenance analysis ideas: According to the display part of the measurement table in turn measurement, found LCD_CABC to ground short circuit, after replacement of U900 fault repair.

3.8 8 Frequency fault

Redmi 6 Pro audio circuit includes speaker, microphone, earpiece and earphone. After the audio signals come out of U1100, they go to different channels respectively. When analyzing this kind of fault, first distinguish which part has the problem according to the fault phenomenon, and then analyze and repair according to the following respective modules.

Principle block diagram of audio signal part:



3.8.1 1 speaker failure

Speaker is connected to the motherboard through FPC, and the speaker has a separate power amplifier. U2005 audio signal is processed by U100, and the power amplifier is controlled by enabling signal. The signal direction is as follows: the signal is processed by CPU and then goes to CODEC, then amplified by audio power amplifier U2005, connected to the sub-board through interface J2301, and connected to the speaker through ANT 102 and ANT 101 on the sub-board.

Maintenance analysis ideas:

1. Visually check whether J2301 looks good.
2. Software upgrades troubleshoot software.
3. Measure whether SPK + and SPK - voltage are normal.
4. Measure whether SPK_SENSE_P and SPK_SENSE_N are normal to ground.
4. If the above signals are normal, consider whether the bus from CODEC to CPU is normal and replace U100. Maintenance Case 1

Fault Phenomenon:

Speaker Silent Fault

Component: L2020

Maintenance analysis idea: L2020 is damaged, resulting in AUDIO PA U2005 without VPH_PWR power supply, and the fault is repaired after replacement.

3.8.2 2 MIC fault

Redmi 6 Pro contains two MIC circuits, the main/auxiliary MIC and the auxiliary wheat are on the main board, and the main MIC is welded on the auxiliary board. First, distinguish which part has the problem according to the fault phenomenon, and then analyze and repair according to the following modules.

Maintenance analysis ideas:

1. Visually check whether J2301 looks good.
2. Measure whether MIC +, MIC - and MIC_BIAS1 are normal to ground.
3. Measure whether the MIC_BIAS1 voltage is normal.
4. If the above signals are normal, consider whether the bus between CODEC and CPU is normal.

3.8.3 3 earpiece failure

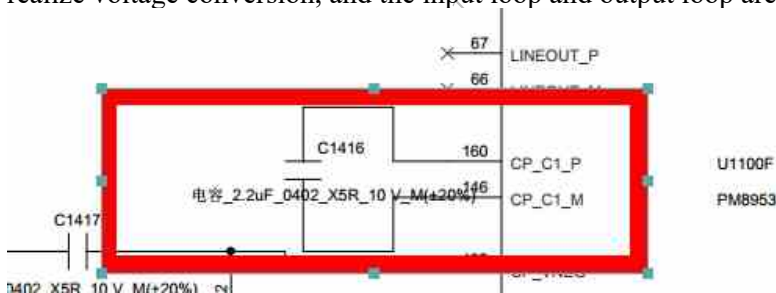
Maintenance analysis ideas:

1. Check whether the motherboard PAD2001 and PAD2002 have bad appearance.
2. Software upgrade, troubleshooting software.
3. Measure whether the receiver path is normal, and whether the resistance values of PAD2001 and PAD2002 are abnormal.
4. According to the audio signal trend of REC measured in the circuit diagram, reverse analysis is carried out.

3.8.4 Machine 4 failure

Maintenance analysis ideas:

1. Check J2000 for foreign body deformation.
2. Software upgrade, troubleshooting software.
3. Measure whether the earphone DET detection signal has 1.8 V voltage.
4. According to the working conditions of each component in the earphone channel, the relevant signals are measured step by step. If the U100 is replaced normally.
5. Check whether the charging pump circuit is normal. Charge pump is the output of the sound source in the left and right channels of headphones. Charge pump uses the charging and discharging of capacitors to realize voltage conversion, and the input loop and output loop are turned on in turn.



6. Turn on the audio plug-in headset test to see if CDC has audio output. If the output signal is abnormal, focus on detecting whether the CDC_HPH_REF to the earphone interface FM_ANT in the figure is open.

Maintenance Case 1

Failure phenomenon: Do not identify the cause of headphone failure:

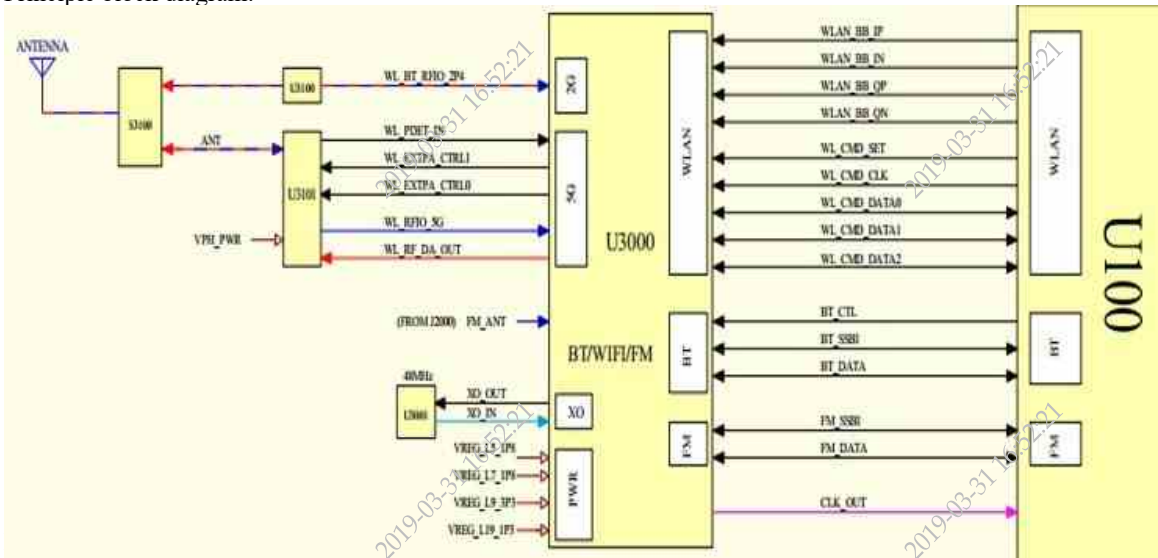
software failure

Maintenance analysis idea: There is no abnormality in appearance, try to repair the fault after brushing the machine.

3.9 9 WIFI/BT/FM Failure

Principle: WIFI/BT/FM is integrated in U3000, CPU realizes communication and control to U3000 through bus, and U1100 provides power supply to U3000. BT and WL still use common RF path, and signals are received through antenna J3100, filtered by U3100 after S3100 frequency selection, and then entered U3000.

Principle block diagram:



FM、BT、WIFI测量表

Symbol	测量值	测量点
VREG_L7_1P8	1.8V	C3005
VREG_L5_1P8	1.8V	C3004
CLK_OUT	48MHz	R3001
VREG_L9_3P3	3.3V	R3002
VREG_L19_1P3	1.3V	R3003

1. Software upgrade, troubleshooting software.
2. Measure whether U3001 is normal (48MHz).
3. Measure whether the power supply, clock and enable signals of U3000 are normal.
4. Remove U3000 and measure whether the bus between U100 is normal. If it is normal, replace U3000.
5. Replace U100.

Maintenance Case 1

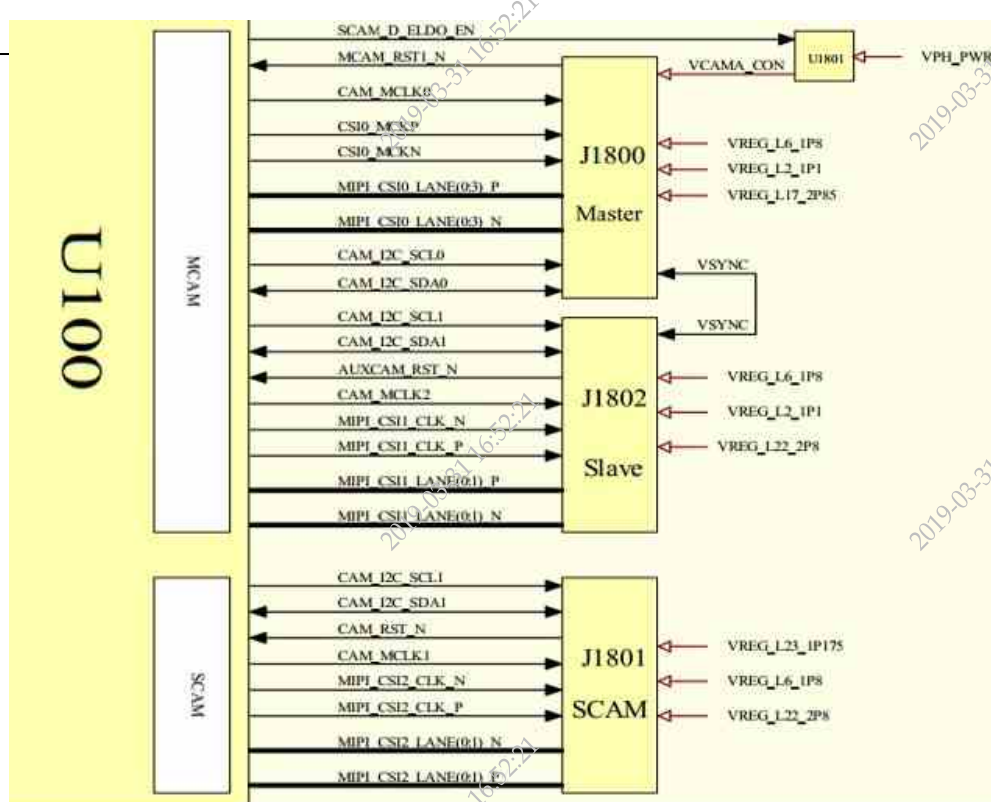
Fault Phenomenon: No
Data Transmission Fault
Component: U100

Maintenance analysis ideas: Wifi can be opened with connection but no data transmission, trying to brush the machine is invalid, and the fault is repaired after replacing U100.

3.10 0 Camera failure

Principle introduction: The front Camera and the rear Camera have two common power supply signals, VREG_L6_1P8 and VREG_L22_2P8. If the front and rear cameras can't be opened, these two power supply signals should be given priority.

Principle block diagram of camera circuit:



Front camera signal measuring table:

Sub Cam关键信号	测量值	测量点	条件
VREG_L23_1P175	1.75V	R1821	开启前摄测量
VREG_L6_1P8	1.8V	C1816	
VREG_L22_2P8	2.8V	R1807	开启前摄测量
CAM_MCLK1	1V	R1804	开启前摄测量

Main Cam关键信号	测量值	测量点	条件
VREG_L6_1P8	1.8V	R1801	
VREG_L2_1P1	1.1V	R1803	开启后摄测量
VREG_L17_2P85	2.85V	C1805	开启后摄测量
CAM_MCLK0	1V	R1809	开启后摄测量
VREG_L22_2P8	2.8V	R1823	开启后摄测量
CAM_MCLK2	1V	R1822	开启后摄测量

Maintenance analysis ideas:

1. Detect J1801, J1802 and J1800 and surrounding components for missing or damaged.
2. Software upgrade, troubleshooting software.
3. Go to CIT to test the front camera and the rear camera to distinguish faults.
4. Measure whether the ground values of J1801, J1802 and J1800 are normal.

Main Camera J1800														
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
710	GND	520	GND	520	520	∞	550	GND	510	510	GND	510	GND	520
GND	792	792	GND	804	804	GND	785	785	GND	780	780	GND	792	792
30	29	28	27	26	25	24	23	22	21	20	19	18	17	16

Front Camera J1801											
1	2	3	4	5	6	7	8	9	10	11	12
788	788	GND	792	805	GND	794	795	GND	∞	∞	GND
517	∞	508	508	521	452	549	GND	∞	∞	GND	518
24	23	22	21	20	19	18	17	16	15	14	13

AUX Camera J1800Z												
28	1	2	3	4	5	6	7	8	9	10	11	12
GND	GND	519	GND	711	GND	512	∞	510	510	GND	629	GND
GND	GND	792	792	GND	800	811	GND	797	797	GND	455	GND
25	24	23	22	21	20	19	18	17	16	15	14	13

5. Measure whether the camera power supply, clock and reset signal output are normal.
6. Measure whether I2C and MIPI bus of U100 output are normal.

Maintenance Case 1

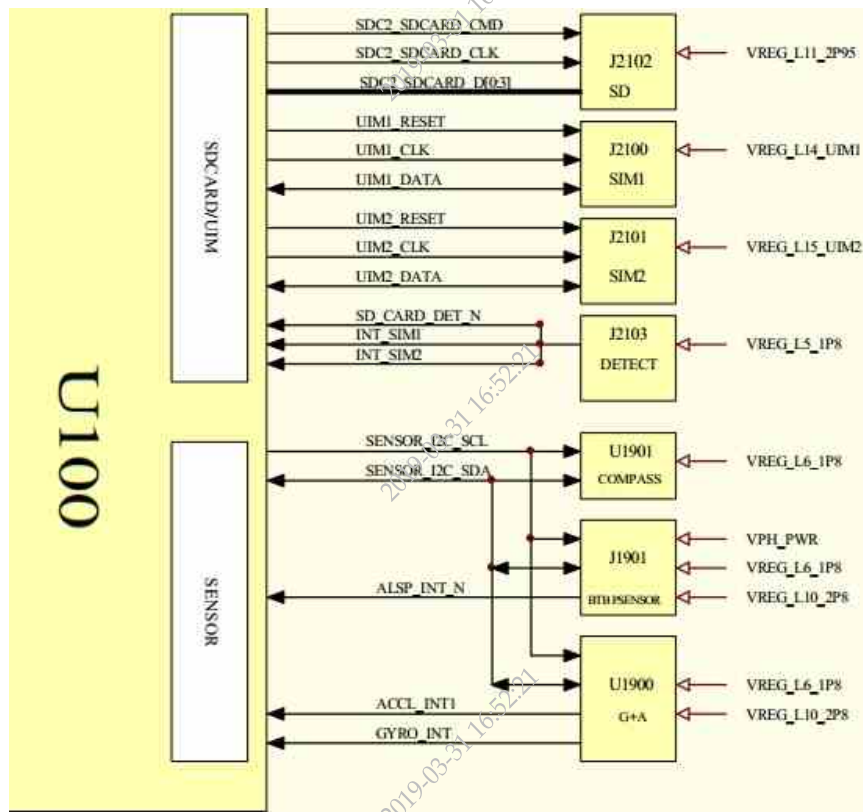
Failure phenomenon:
The rear camera can't be
opened. Failure reason:
U100

Maintenance analysis idea: Measure E1802 MIPI CSI 0_LANE3_N with no value at both ends, remove E1802 fault, measure one end connecting U100 with no value after removing E1802, and repair the fault after replacing U100.

3.11 1 sensor failure

The main sensors in Xiaomi 5X are U1901 (M-SENSOR electronic compass), J1900 (light-distance sensor interface) and U1900 (gravity-gyroscope sensor).

Principle block diagram of inductor circuit:



传感器测量表			
元件名称	Symbol	测量值	测量点
共用信号	SENSOR_I2C_SDA	1.8V	R301
	SENSOR_I2C_SCL	1.8V	R302
U1901	VREG_L6_1P8	1.8V	C1901
J1900	VPH_PWR	4V	R1911
	VREG_L10_2P8	2.8V	C1914
	VREG_L6_1P8	1.8V	R1905
U1900	VREG_L6_1P8	1.8V	C1904
	VREG_L10_2P8	2.8V	C1902

1. Check the appearance of corresponding sensors and interfaces to eliminate appearance faults.
2. Software upgrade, troubleshooting software.
3. Measure whether the working conditions of the corresponding sensors are normal.
4. Replace the corresponding sensor and U100.

Maintenance Case 1

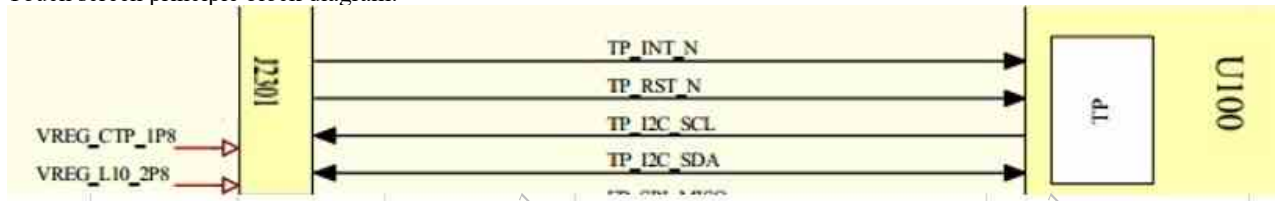
Failure phenomenon:
Compass failure Failure
cause: U1901

Maintenance analysis idea: Geomagnetic test, however, visual inspection found that the edge of U1901 was tilted, and the fault was repaired after replacement.

3.12 2 touch screen failure

The touch screen is connected with the motherboard by J2301 interface and controlled by U100. Touch screen circuit signals include power supply, INT, RST and I2C, and these signals can directly measure their voltage without connecting with the touch screen.

Touch screen principle block diagram:



Touch screen voltage meter:

触屏测量表		
Symbol	测量值	测量点
VREG_CTP_1P8	1.8V	C1705
TP_INT_N	1.8V	J2301#4
TP_I2C_SDA	1.8V	TP2302
TP_I2C_SCL	1.8V	TP2301
TP_RST_N	1.8V	R2313
VREG_L10_2P8	2.8V	C1308

1. Check J2301 and surrounding components for damage.
2. Software upgrade, troubleshooting software.
3. Check whether the signal of touch screen meter is normal.
4. Replace U100.

Maintenance Case 1

Fault phenomenon:

touch screen failure

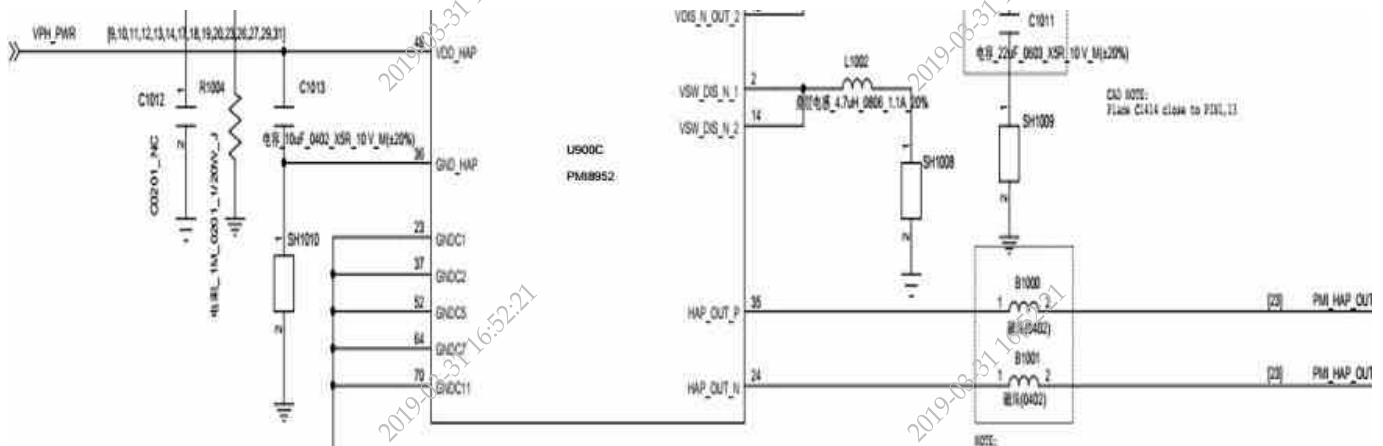
Fault component:

J2301

Maintenance analysis ideas: touch screen failure, visual inspection interface J2301 found that the 3rd and 4th pin pins were deformed, and the failure was repaired after replacing new components.

3.13 3 vibration related function failure

Schematic diagram:

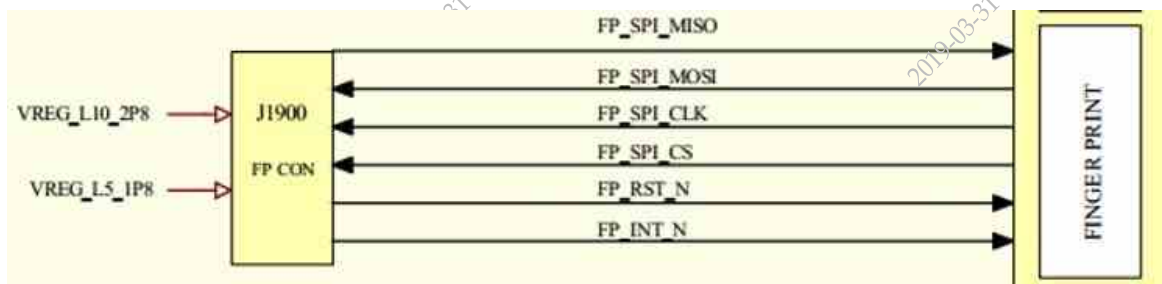


Maintenance analysis ideas:

1. Software upgrade, troubleshooting software.
2. Measure whether PMI_HAP_OUT_P and PMI_HAP_OUT_N signals are normal. The connection points between the sub-board and the vibrator are ANT 104 and ANT 105. If the signals are abnormal, replace U900 after eliminating the component faults on its path respectively.
3. Measure whether the control signal HAPT_PWM_LINE_IN is normal, and replace U100 if it is abnormal.

3.14 4 Fingerprint identification failure

Fingerprint principle block diagram:



Fingerprint voltage meter:

指纹测量表		
Symbol	测量值	测量点
VREG_L5_1P8	1.8V	R1908
VREG_L10_2P8	2.8V	R1907
FP_RST_N	1.8V	J1900第6脚
FP_SPI_CS	1.8V	J1900第9脚
FP_SPI_CLK	480	J1900第10脚
FP_SPI_MISO	480	J1900第8脚
FP_SPI_MOSI	480	J1900第7脚
FP_INT_N	1.8V	J1900第5脚

1. Visually check whether J1900 and its surrounding components are damaged, and replace them if they are damaged.
2. Brush the machine to eliminate software faults.
3. Measure whether the above voltage and other signals are normal.
4. If all the above signals are normal, replace U100.